PRACTICAL 4 - UNDERSTANDING DATA WITH GRAPHS

Objective

The objective of this practical is to understand the TeachingRatings dataset using descriptive statistics and graphical methods. The analysis includes identifying duplicate observations, comparing teaching evaluations across course divisions, and visualizing the relationship between professor age, teaching evaluation, gender, and tenure.

Dataset Information

The dataset contains 463 observations and 19 variables. The variable 'prof' identifies individual professors, with 94 unique professors in the dataset.

Q.1) Identify all duplicate cases using prof. Using all observations, find the average and standard deviation for age. Repeat the analysis by first filtering the dataset to include one observation for each instructor with a total number of observations restricted to 94.

Ans: The mean age decreases slightly after restricting the dataset to one observation per professor, while the standard deviation increases slightly. This shows that repeated observations for the same professor can influence descriptive statistics.

Input / Code Screenshot

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# ============================================================

# QUESTION 1

# ============================================================



print("\n" + "=" * 70)

print("QUESTION 1")

print("=" * 70)



# Identify duplicate cases using professor ID

duplicate_cases = ratings_df[

   ratings_df.duplicated(subset=["prof"], keep=False)

].sort_values("prof")
```

```
print("\nDuplicate cases based on professor ID:")
print(duplicate_cases[["prof", "age", "gender", "eval", "tenure"]])

# Number of observations and unique professors
total_observations = len(ratings_df)
unique_professors = ratings_df["prof"].nunique()

print("\nTotal observations:", total_observations)
print("Number of unique professors:", unique_professors)

# Age statistics using all observations
all_mean_age = ratings_df["age"].mean()
all_std_age = ratings_df["age"].std()

print("\nUsing all observations:")
print("Mean age:", round(all_mean_age, 4))
print("Standard deviation of age:", round(all_std_age, 4))

# Keep only one observation for each professor
one_per_professor = ratings_df.drop_duplicates(
   subset=["prof"],
   keep="first"
)

filtered_mean_age = one_per_professor["age"].mean()
filtered_std_age = one_per_professor["age"].std()

print("\nAfter keeping one observation per professor:")
print("Number of observations:", len(one_per_professor))
print("Mean age:", round(filtered_mean_age, 4))
print("Standard deviation of age:", round(filtered_std_age, 4))
```

Output Screenshot

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======================================================================
QUESTION 1
======================================================================

Duplicate cases based on professor ID:
     prof  age  gender  eval tenure
0       1   36  female   4.3    yes
1       1   36  female   3.7    yes
2       1   36  female   3.6    yes
3       1   36  female   4.4    yes
4       2   59    male   4.5    yes
..    ...  ...     ...   ...    ...
450    93   32    male   3.2    yes
453    94   42  female   3.3    yes
454    94   42  female   3.2    yes
452    94   42  female   3.3    yes
455    94   42  female   4.1    yes

[456 rows x 5 columns]

Total observations: 463
Number of unique professors: 94

Using all observations:
Mean age: 48.365
Standard deviation of age: 9.8027

After keeping one observation per professor:
Number of observations: 94
Mean age: 47.5532
Standard deviation of age: 10.2565
```

Q.2) Using a bar chart, demonstrate if instructors teaching lower-division courses receive higher average teaching evaluations.

Ans: Yes. Instructors teaching lower-division courses receive a higher average teaching evaluation (4.0873) than instructors teaching upper-division courses (3.9526).

Input / Code Screenshot

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# ============================================================

# QUESTION 2

# ============================================================


print("\n" + "=" * 70)

print("QUESTION 2")

print("=" * 70)
```

```
# Calculate average evaluation for each division

division_evaluation = (

   ratings_df

   .groupby("division")["eval"]

   .mean()

)


print("\nAverage evaluation by course division:")

print(division_evaluation.round(4))


# Create bar chart

plt.figure(figsize=(8, 5))


division_evaluation.plot(

   kind="bar",

   edgecolor="black"

)


plt.xlabel("Course Division")

plt.ylabel("Average Teaching Evaluation")

plt.title("Average Teaching Evaluation by Course Division")

plt.xticks(rotation=0)


plt.tight_layout()


plt.savefig(

   "output/q2_average_evaluation_by_division.png",

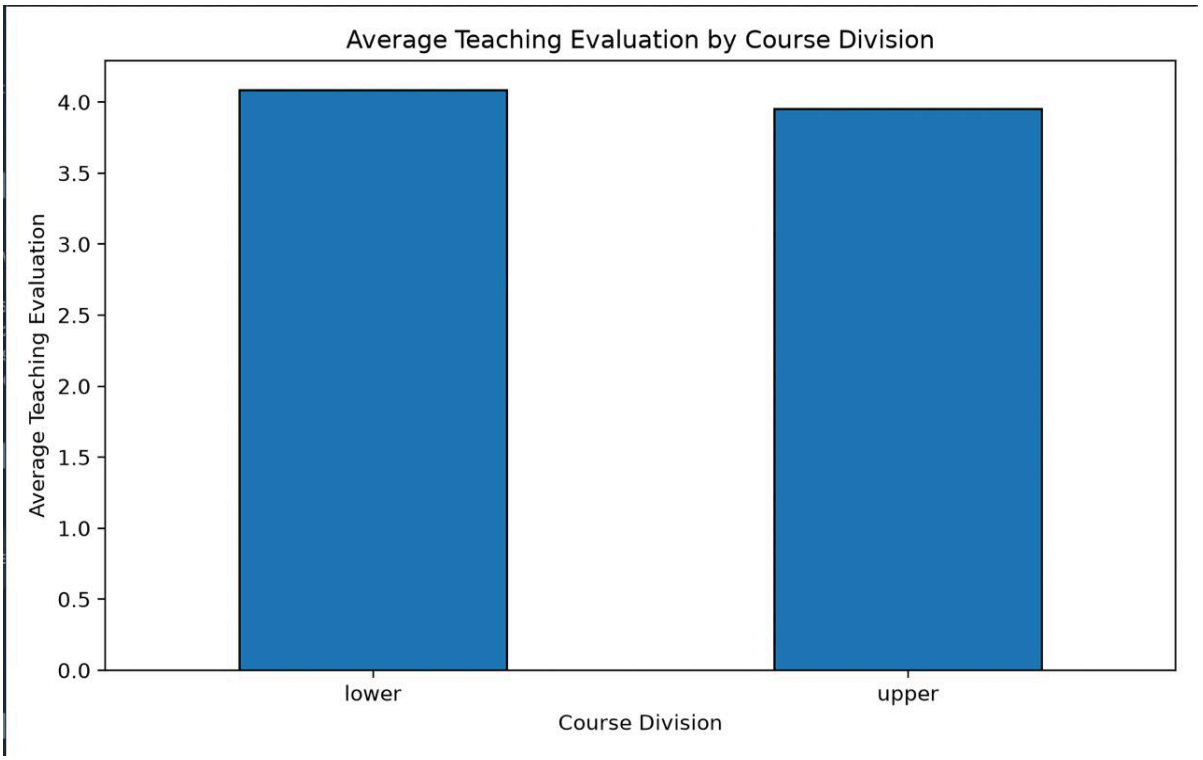
   dpi=300,

   bbox_inches="tight"

)


plt.show()
```

Output 2



Q.3) Plot the relationship between age and teaching evaluation scores.

Ans: The scatter plot does not show a strong clear linear relationship between age and teaching evaluation scores. Professors of different ages can have both high and low evaluation scores.

Input / Code Screenshot

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# ============================================================

# QUESTION 3

# ============================================================


print("\n" + "=" * 70)

print("QUESTION 3")

print("=" * 70)


print("\nPlotting age against teaching evaluation scores...")
```

```
plt.figure(figsize=(8, 5))


plt.scatter(
   ratings_df["age"],
   ratings_df["eval"],
   alpha=0.7
)

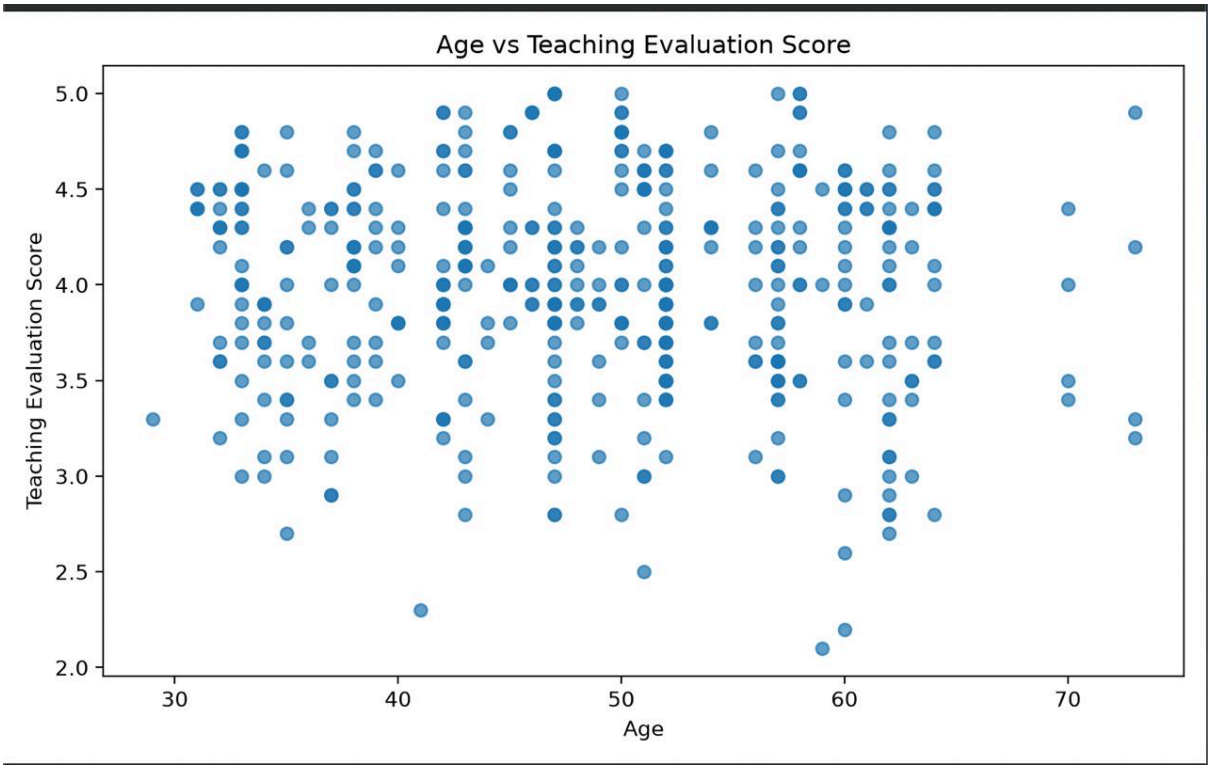

plt.xlabel("Age")
plt.ylabel("Teaching Evaluation Score")
plt.title("Age vs Teaching Evaluation Score")


plt.tight_layout()


plt.savefig(
   "output/q3_age_vs_evaluation.png",
   dpi=300,
   bbox_inches="tight"
)


plt.show()
```

Output 3



Q.4) Using gender-differentiated scatter plots, plot the relationship between age and teaching evaluation scores.

Ans: There is no obvious strong age-evaluation pattern that is clearly different between the two genders.

Input / Code Screenshot

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# ============================================================

# QUESTION 4

# ============================================================


print("\n" + "=" * 70)

print("QUESTION 4")

print("=" * 70)


print("\nPlotting age against evaluation by gender...")


plt.figure(figsize=(8, 5))
```

```
for gender in sorted(ratings_df["gender"].dropna().unique()):


    gender_data = ratings_df[

        ratings_df["gender"] == gender

    ]


    plt.scatter(

        gender_data["age"],

        gender_data["eval"],

        alpha=0.7,

        label=gender

    )


plt.xlabel("Age")

plt.ylabel("Teaching Evaluation Score")

plt.title("Age vs Teaching Evaluation by Gender")

plt.legend(title="Gender")


plt.tight_layout()


plt.savefig(

    "output/q4_age_vs_evaluation_by_gender.png",

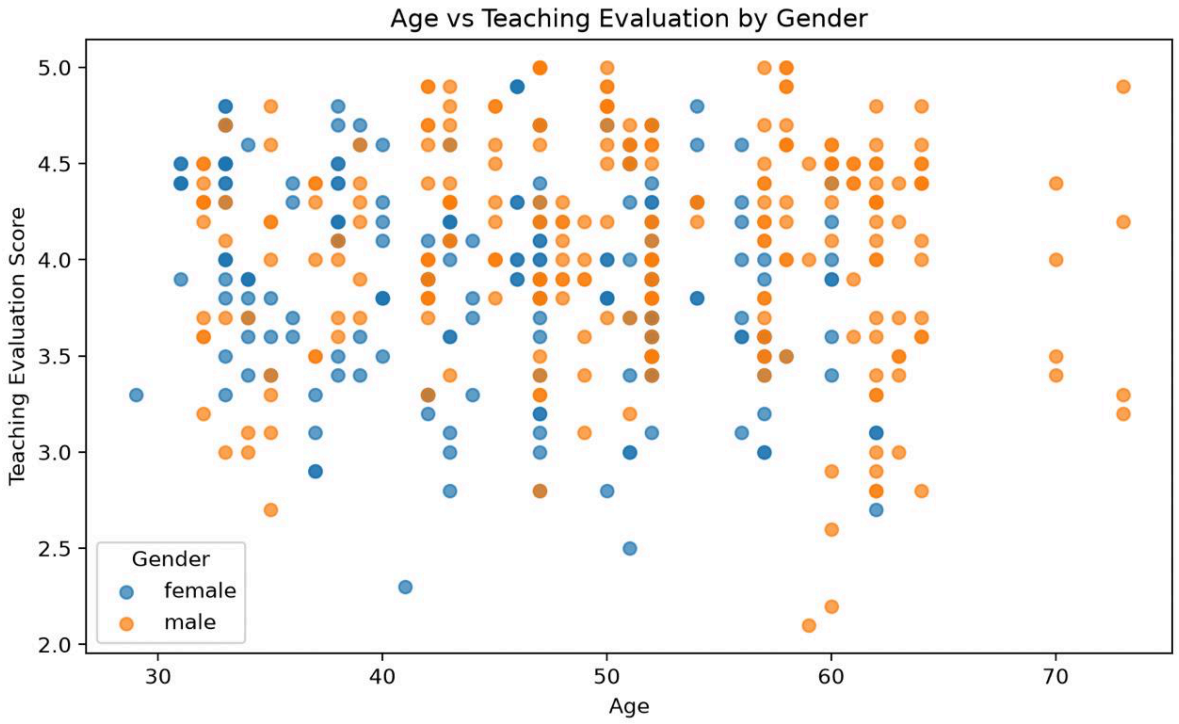
    dpi=300,

    bbox_inches="tight"

)


plt.show()
```

Output 4



Q.5) Create a scatterplot of age and evaluation scores, differentiated by gender and tenure.

Ans: The four groups show substantial overlap in age and evaluation scores. Overall, the plot does not indicate a simple linear relationship between age and evaluation score.

Input / Code Screenshot

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# ============================================================

# QUESTION 5

# ============================================================


print("\n" + "=" * 70)

print("QUESTION 5")

print("=" * 70)


print("\nPlotting age against evaluation by gender and tenure...")


plt.figure(figsize=(9, 6))
```

```
for gender in sorted(ratings_df["gender"].dropna().unique()):

   for tenure in sorted(ratings_df["tenure"].dropna().unique()):

       group_data = ratings_df[
           (ratings_df["gender"] == gender) &
           (ratings_df["tenure"] == tenure)
       ]

       if len(group_data) > 0:

           plt.scatter(
               group_data["age"],
               group_data["eval"],
               alpha=0.7,
               label=f"{gender}, tenure={tenure}"
           )

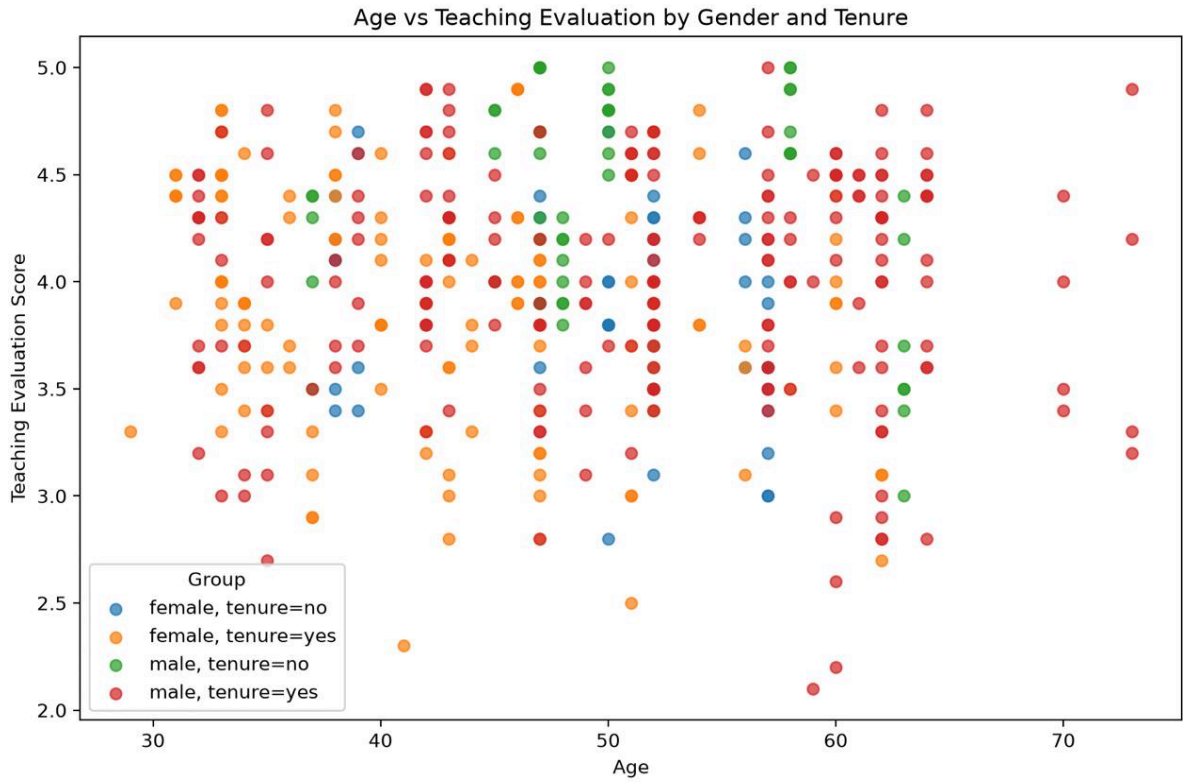
plt.xlabel("Age")
plt.ylabel("Teaching Evaluation Score")
plt.title("Age vs Teaching Evaluation by Gender and Tenure")
plt.legend(title="Group")

plt.tight_layout()

plt.savefig(
   "output/q5_age_vs_evaluation_gender_tenure.png",
   dpi=300,
   bbox_inches="tight"
)

plt.show()
```

Output 5



CONCLUSION:

The TeachingRatings dataset was analyzed using descriptive statistics and graphical techniques. Duplicate observations were identified using professor ID, age statistics were calculated before and after restricting the data to one observation per professor, and graphical analysis was performed for course division, age, gender, and tenure. Lower-division courses had the higher average teaching evaluation. The scatter plots did not show a strong obvious linear relationship between age and teaching evaluation.

Submitted by

Name: Anjani Bharat

Reg. No.: PGD202563026

Submitted to

Faculty: Anitya Kumar Gupta